

Challenge (Habanero)

Q1. Expand $(x + 5)^2$ using special patterns

- (A) $x^2 + 10x + 25$
 (B) $x^2 - 10x + 25$
 (C) $x^2 + 5x - 25$
 (D) $x^2 - 5x + 25$
 (E) $x^2 + 25x - 5$

(E) $x^2 - 4x - 4$

Q2. What is the factored form of $x^2 + 12x + 36$?

- (A) $(x + 6)^2$
 (B) $(x - 6)^2$
 (C) $(x + 3)^2$
 (D) $(x - 3)^2$
 (E) $(x + 12)^2$

Q3. A construction worker is building a rectangular foundation with a length of $(x+4)$ meters and a width of $(x-2)$ meters. What is the area of the foundation?

- (A) $x^2 + 2x - 8$
 (B) $x^2 - 2x - 8$
 (C) $x^2 + 4x - 2$
 (D) $x^2 - 4x - 2$
 (E) $x^2 + 2x + 8$

Q4. A recipe calls for $(x + 3)$ cups of flour and $(x - 1)$ cups of sugar. What is the total amount of ingredients needed?

- (A) $2x + 2$
 (B) $2x - 2$
 (C) $x + 2$
 (D) $x - 2$
 (E) $2x + 1$

Q5. What is the expanded form of $(x - 2)^2$?

- (A) $x^2 - 4x + 4$
 (B) $x^2 + 4x + 4$
 (C) $x^2 - 2x + 4$
 (D) $x^2 + 2x - 4$

Q6. A carpenter is building a triangular roof with a base of $(x + 1)$ meters and a height of $(x - 3)$ meters. What is the area of the roof?

(A) $\frac{1}{2}12x^2 - \frac{1}{2}12x - \frac{1}{2}32$

(A) $\frac{1}{2}12x^2 + \frac{1}{2}12x - \frac{1}{2}32$

(A) $\frac{1}{2}12x^2 - \frac{1}{2}32x + \frac{1}{2}12$

(A) $\frac{1}{2}12x^2 + \frac{1}{2}32x - \frac{1}{2}12$

(A) $x^2 - 2x - 3$

Q7. What is the factored form of $x^2 - 14x + 49$?

- (A) $(x - 7)^2$
 (B) $(x + 7)^2$
 (C) $(x - 2)^2$
 (D) $(x + 2)^2$
 (E) $(x - 1)^2$

Q8. A chef is making a circular cake with a radius of $(x + 2)$ centimeters. What is the area of the cake?

- (A) $\pi(x + 2)^2$
 (B) $\pi(x - 2)^2$
 (C) $\pi x^2 + 4\pi x + 4\pi$
 (D) $\pi x^2 - 4\pi x + 4\pi$
 (E) $\pi x^2 + 2\pi x - 4\pi$

Open Ended Questions (FRQ)

Q9. A contractor is building a rectangular house with a length of $(x + 5)$ meters and a width of $(x - 3)$ meters. Find the area and perimeter of the house.

Q10. A baker is making a batch of cookies using a recipe that calls for $(x + 2)$ cups of flour and $(x - 1)$ cups of sugar. If the baker wants to make 5 batches of cookies, find the total amount of flour and sugar needed.

Chef's Tips

- Check for correct units and measurements
- Ensure students show all work and calculations
- Emphasize the importance of following recipes and instructions carefully